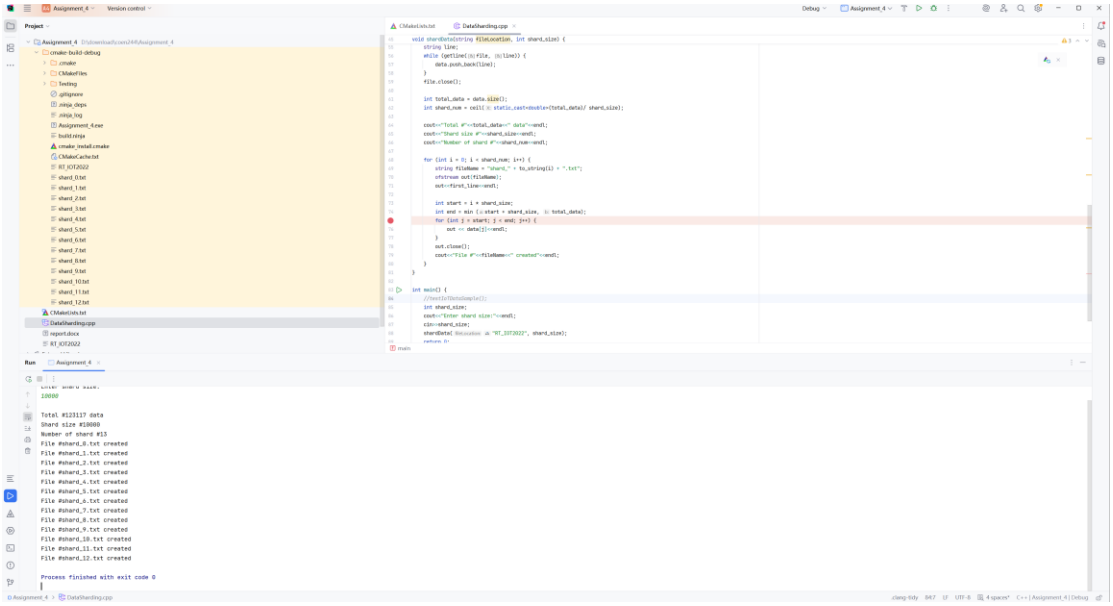
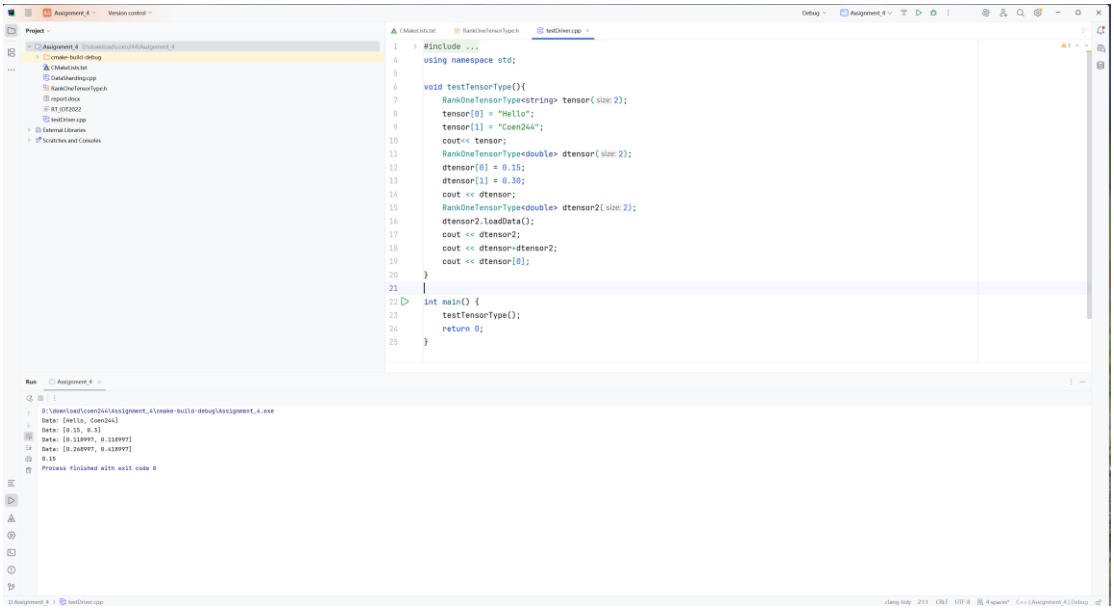
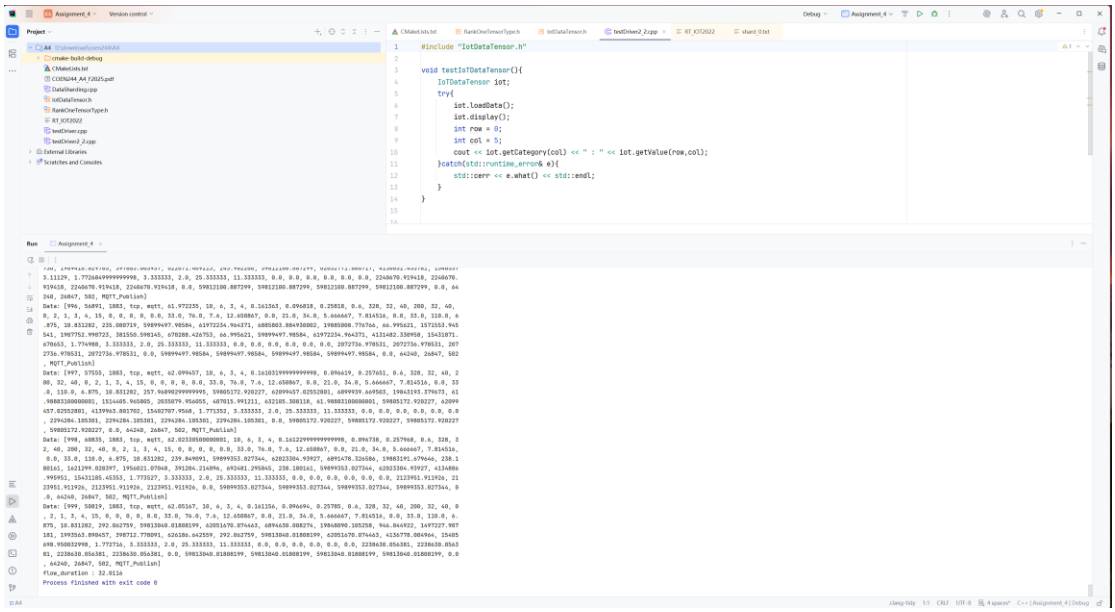
In Task 1, the RT_IOT2022 file contained a total of 123,118 lines. With a shard size of 10,000 entered, a total of 13 shard files were generated, ranging from shard_0 to shard_12. The following screenshot displays the console output. These shard files are contained within the cmake-build-debug folder.



In Task 2.1, the template function with operator overloading was programmed. And loadData() can generate random double floating-point numbers within the range 0 to 1. The following screenshot displays the console output.



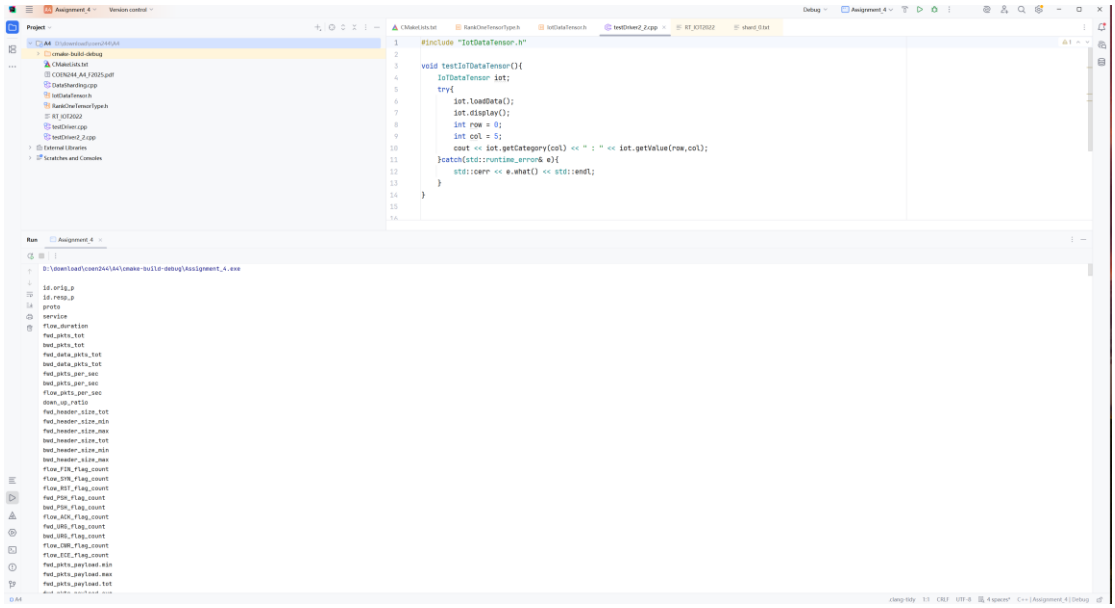
In Task 2.2, RT_IOT2022 is redivided into shard files of size 1000. Running testDriver2_2 prints the names of each category in shard_0.txt, then outputs 1000 data (0-999), and finally outputs the data from the first row, sixth column (row = 0, col = 5). The following screenshots display the console output.



```
#include "IotDataTensor.h"

void testIoTDataTensor(){
    IoTDataTensor iot;
    try{
        iot.loadData();
        iot.display();
        int row = 0;
        int col = 5;
        cout << iot.getCategory(col) << " : " << iot.getValue(row,col);
    }catch(std::runtime_error& e){
        std::cerr << e.what() << std::endl;
    }
}
```

```
3.11129, 1.7726849999999998, 3.333333, 2.0, 25.333333, 11.333333, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2240670.919418, 2240670.
919418, 2240670.919418, 2240670.919418, 0.0, 59812100.887299, 59812100.887299, 59812100.887299, 59812100.887299, 0.0, 64
240, 26847, 502, MQTT_Publish]
Date: [996, 56891, 1883, tcp, mqtt, 61.972235, 18, 6, 3, 4, 0.163363, 0.096818, 0.25818, 0.6, 328, 32, 40, 200, 32, 40,
0, 2, 1, 3, 4, 15, 0, 0, 0, 0, 0.0, 33.0, 70.0, 7.6, 12.658867, 0.0, 21.0, 34.0, 5.666667, 7.814516, 0.0, 33.0, 118.0, 6
.875, 18.831282, 255.000719, 59899497.98584, 63972234.964371, 68858863.884938002, 198858888.778764, 66.995621, 1572153.945
541, 1987752.998723, 381550.5981145, 670288.426753, 66.995621, 59899497.98584, 61972234.964371, 4131482.330958, 15431871.
670863, 1.774988, 3.333333, 2.0, 25.333333, 11.333333, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2072736.978531, 2072736.978531, 207
2736.978531, 2072736.978531, 0.0, 59899497.98584, 59899497.98584, 59899497.98584, 59899497.98584, 0.0, 64240, 26847, 502
, MQTT_Publish]
Date: [997, 57555, 1883, tcp, mqtt, 62.099407, 18, 6, 3, 4, 0.16383199999999998, 0.096619, 0.257651, 0.6, 328, 32, 40, 2
00, 32, 40, 0, 2, 1, 3, 4, 15, 0, 0, 0, 0, 0.0, 33.0, 70.0, 7.4, 12.658867, 0.0, 21.0, 34.0, 5.666667, 7.814516, 0.0, 33
.0, 118.0, 6.875, 18.831282, 257.9689029999999, 59885172.920227, 62899457.02552801, 6899939.669503, 19843193.379673, 61
.9888310000000001, 1514485.965805, 2035079.956055, 487015.991211, 632185.308118, 61.9888310000000001, 59885172.920227, 62099
457.02552801, 6119965.8017702, 154027097.9568, 1.771332, 3.333333, 2.0, 25.333333, 11.333333, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
, 2294286.185301, 2294286.185301, 2294286.185301, 2294286.185301, 0.0, 59885172.920227, 59885172.920227, 59885172.920227
, 59885172.920227, 0.0, 64240, 26847, 502, MQTT_Publish]
Date: [998, 60835, 1883, tcp, mqtt, 62.023385000000001, 18, 6, 3, 4, 0.16122999999999998, 0.096738, 0.257948, 0.6, 328, 3
2, 40, 200, 32, 40, 0, 2, 1, 3, 4, 15, 0, 0, 0, 0, 0.0, 33.0, 70.0, 7.6, 12.658867, 0.0, 21.0, 34.0, 5.666667, 7.814516,
 0.0, 33.0, 118.0, 6.875, 18.831282, 239.849091, 59899353.027344, 62823304.93927, 6891478.32b586, 19883191.679646, 238.1
80161, 1621299.028397, 1956821.07048, 391286.214096, 692481.295845, 238.180161, 59899353.027344, 62823304.93927, 4134886
.9959561, 15431285.463353, 1.773527, 3.333333, 2.0, 25.333333, 11.333333, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2123951.911926, 21
23951.911926, 2123951.911926, 2123951.911926, 0.0, 59899353.027344, 59899353.027344, 59899353.027344, 59899353.027344, 0
.0, 64240, 26847, 502, MQTT_Publish]
Date: [999, 59819, 1883, tcp, mqtt, 62.05167, 18, 6, 3, 4, 0.161156, 0.096694, 0.25785, 0.6, 328, 32, 40, 200, 32, 40, 0
, 2, 1, 3, 4, 15, 0, 0, 0, 0, 0.0, 33.0, 70.0, 7.6, 12.658867, 0.0, 21.0, 34.0, 5.666667, 7.814516, 0.0, 33.0, 118.0, 6.
875, 18.831282, 292.862759, 59813848.01888199, 62851670.8744a5, 6896438.088274, 19848090.185258, 946.844922, 1497227.987
181, 1993563.8989457, 398712.778091, 624186.642559, 292.862759, 59813848.01888199, 62851670.8744a5, 4156778.864964, 15485
698.9508329998, 1.772716, 3.333333, 2.0, 25.333333, 11.333333, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 2238630.856381, 2238630.8563
81, 2238630.856381, 2238630.856381, 0.0, 59813048.01888199, 59813848.01888199, 59813848.01888199, 59813848.01888199, 0.0
, 64240, 26847, 502, MQTT_Publish]
flow_duration : 32.011s
Process finished with exit code 0
```



```
#include "IotDataTensor.h"

void testIoTDataTensor(){
    IoTDataTensor iot;
    try{
        iot.loadData();
        iot.display();
        int row = 0;
        int col = 5;
        cout << iot.getCategory(col) << " : " << iot.getValue(row,col);
    }catch(std::runtime_error& e){
        std::cerr << e.what() << std::endl;
    }
}
```

```
D:\download\csen244\b4\cmake-build-debug\Assignment_4.exe

id.orig_p
id.resp_p
proto
service
flow_duration
fwd_pkts_tot
bwd_pkts_tot
fwd_data_pkts_tot
bwd_data_pkts_tot
fwd_pkts_per_sec
bwd_pkts_per_sec
flow_pkts_per_sec
down_up_ratio
fwd_header_size_tot
fwd_header_size_min
fwd_header_size_max
bwd_header_size_tot
bwd_header_size_min
bwd_header_size_max
flow_FIN_flag_count
flow_SYN_flag_count
flow_RST_flag_count
fwd_PSH_flag_count
bwd_PSH_flag_count
flow_ACK_flag_count
fwd_URG_flag_count
bwd_URG_flag_count
flow_CWR_flag_count
flow_ECE_flag_count
fwd_pkts_payload.min
fwd_pkts_payload.max
fwd_pkts_payload.tot
```